KENNESAW STATE UNIVERSITY
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Smoking During Pregnancy and Preterm Birth: A Doubly Robust Propensity Score Weighting Analysis of Millions of U.S. Births

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PROBLEM STATEMENT

- **Why it matters.** Smoking in pregnancy is a known risk factor for preterm birth, delivery before 37 weeks.
- **The problem.** Mothers who smoke also differ in education, age and health, which affect preterm birth too, so the raw 5.6-point gap overstates what smoking itself does.
- **The question.** How much of that gap does smoking itself cause?
- **The data.** 3,259,496 U.S. singleton births in 2023; 97,187 to mothers who smoked.
- **The approach.** Augmented Inverse Probability of Treatment Weighting (AIPTW), a doubly robust method that adjusts for differences between smokers and non-smokers and produces a valid estimate if either the smoking model or the birth-outcome model is correct.

METHODS

1. Two models. From ten confounders X, a propensity model e(X) gives each mother's chance of smoking; outcome models μ1(X), μ0(X) give her chance of preterm birth with and without smoking.

2. Cross-fitting. Births are split into 5 folds. Logistic, LASSO (all two-way interactions) and random-forest models are fit on 4 folds and score the 5th, so no birth is scored by a model that saw it.

$$e(X) = P(T = 1 \mid X) \qquad \mu_t(X) = P(Y = 1 \mid T = t, X),\ t = 0, 1$$

$$\hat{\tau} = \frac{1}{n}\sum_i \left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{T_i\,(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - T_i)\,(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{e}(X_i)} \right]$$

$$\text{risk difference} = \hat{\tau}\ \text{(points)}; \quad \text{SE} = \text{sd}(\hat{\varphi}_i)/\sqrt{n}, \quad \hat{\varphi}_i = \text{term in brackets} - \hat{\tau}$$

Note. The first two terms are the outcome-model prediction; the last two correct it with weights 1/e(X), so the estimate is right if either model is right.

3. Uncertainty. Standard error from the formula and from a full-sample bootstrap of 1,000 resamples.

4. Checks. Overlap and covariate balance, alternative adjustment sets, E-value.

RESULTS AND FINDINGS

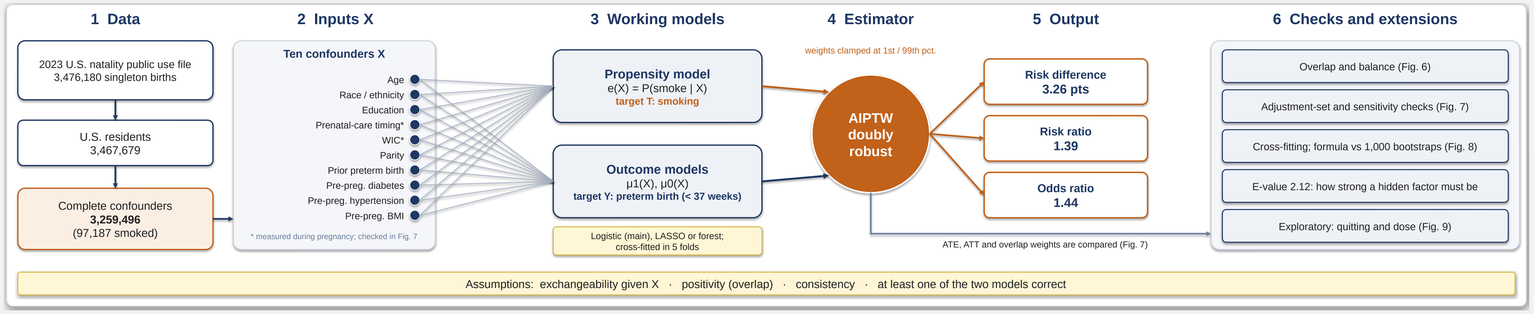
- **Raw gap.** 13.9% of births to mothers who smoked were preterm against 8.3% for others, a 5.6-point gap (Figs. 2, 3).
- **Main estimate.** Adjusted risk difference 3.26 points (bootstrap SE 0.39; 95% interval 2.5 to 4.1). Among mothers who smoked, *about 34 extra preterm births per 1,000* (Fig. 7).
- **Who is compared.** 8.4% of mothers with some high school but no diploma smoked, against 0.1% of those with a doctorate (Fig. 2). Education best separates smokers from others and a prior preterm birth best predicts early delivery; dropping education would move the estimate by +1.03 points (Figs. 4, 5).
- **Balance.** Covariate levels out of balance fall from 21 to 3 after weighting and 0 with overlap weights (Fig. 6).
- **Robustness.** Adjusted estimates stay between 3.0 and 3.7 across analytic choices; LASSO (2.80) and forest (2.30) agree in direction. Formula and bootstrap SEs match (Figs. 7, 8).
- **Sensitivity analysis.** An unmeasured factor would need a risk ratio of about 2.1 with both smoking and preterm birth to explain the result away (E-value).
- **Quitting and dose.** Preterm risk was 8.3% in never-smokers, 9.4% in quitters, 12.5% in mothers who started during pregnancy and 13.9% in those who smoked throughout. After adjustment quitters look like never-smokers (−0.07 points, 95% CI −0.40 to 0.27); the extra risk is similar at every cigarette level (3.3 to 3.6 points). (Fig. 9).

LIMITATIONS

- **Unmeasured confounding.** A strong unmeasured confounder, beyond the E-value above, could still make this study not meaningful.
- **Self-reported smoking** may be under-reported.
- **Limited overlap.** 41% of non-smokers have almost no smoker like them (Fig. 6), so the effect among smokers is sturdier.
- **Care timing and WIC** may follow smoking; without them the estimate is 3.7 (Fig. 7).
- **Observational data;** quit and cigarette results are exploratory and use intake before pregnancy.

RESULTS

After adjustment, mothers who smoked had a 3.3-point higher risk of preterm birth (SE 0.39), a gap two fifths smaller than the raw 5.6 points.



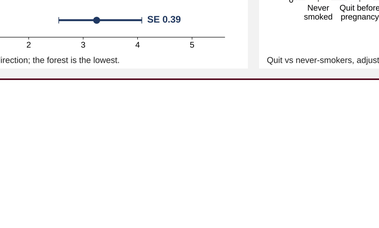
Causal diagram: each confounder affects both

1

Who smokes?

2

% of mothers who smoked, by education

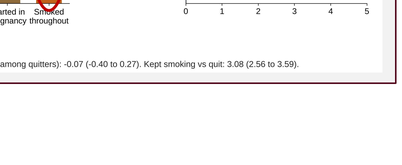


Highlighted: the highest and lowest groups. Mothers cannot be compared across education levels without adjustment.

Smokers deliver earlier

3

% of births at each week of delivery

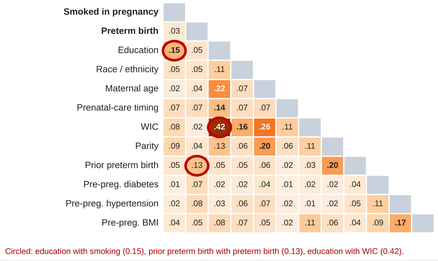


Preterm: 13.9% of smokers vs 8.3% of non-smokers.
Average delivery: 37.97 vs 38.45 weeks.
A previous preterm birth: 8.9% of smokers vs 3.7% of non-smokers,
and it carries a preterm risk of 26.1% (non-smokers) vs 7.6% without it.

How the variables relate

4

Association strength: 0 none, 1 perfect

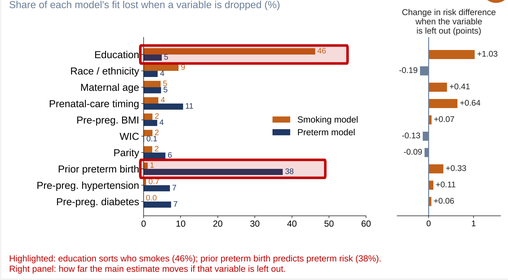


Circled: education with smoking (0.15), prior preterm birth with preterm birth (0.13), education with WIC (0.42).

What drives confounding?

5

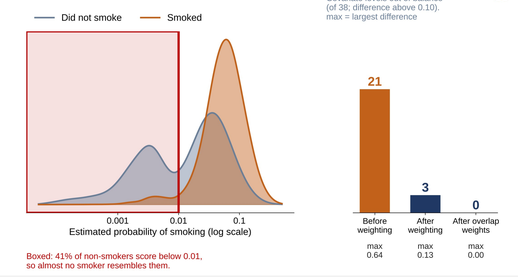
Share of each model's fit lost when a variable is dropped (%)



Highlighted: education sorts who smokes (46%); prior preterm birth predicts preterm risk (38%).
Right panel: how far the main estimate moves if that variable is left out.

Overlap is limited, balance is good

6

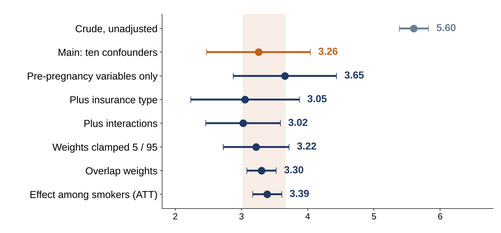


Boxed: 41% of non-smokers score below 0.01, so almost no smoker resembles them.

Result holds across analytic choices

7

Risk difference in preterm birth, percentage points (95% CI)

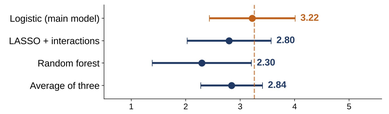


Shaded: adjusted estimates stay between 3.0 and 3.7 points under every choice.

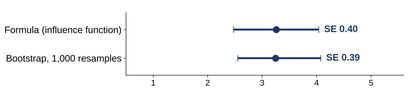
Flexible models and variance

8

Same estimator, cross-fitted learners (95% CI, points)



Standard error: the formula agrees with a full bootstrap

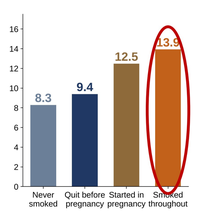


Dashed line: main estimate. Flexible models keep the direction; the forest is the lowest.

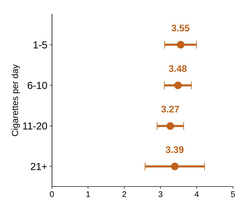
Quitting and cigarettes per day

9

Preterm births (%) by smoking history, crude



Extra risk among mothers who smoked, by cigarettes per day before pregnancy (points, 95% CI)



Quit vs never-smokers, adjusted (among quitters): -0.07 (-0.40 to 0.27). Kept smoking vs quit: 3.08 (2.56 to 3.59).